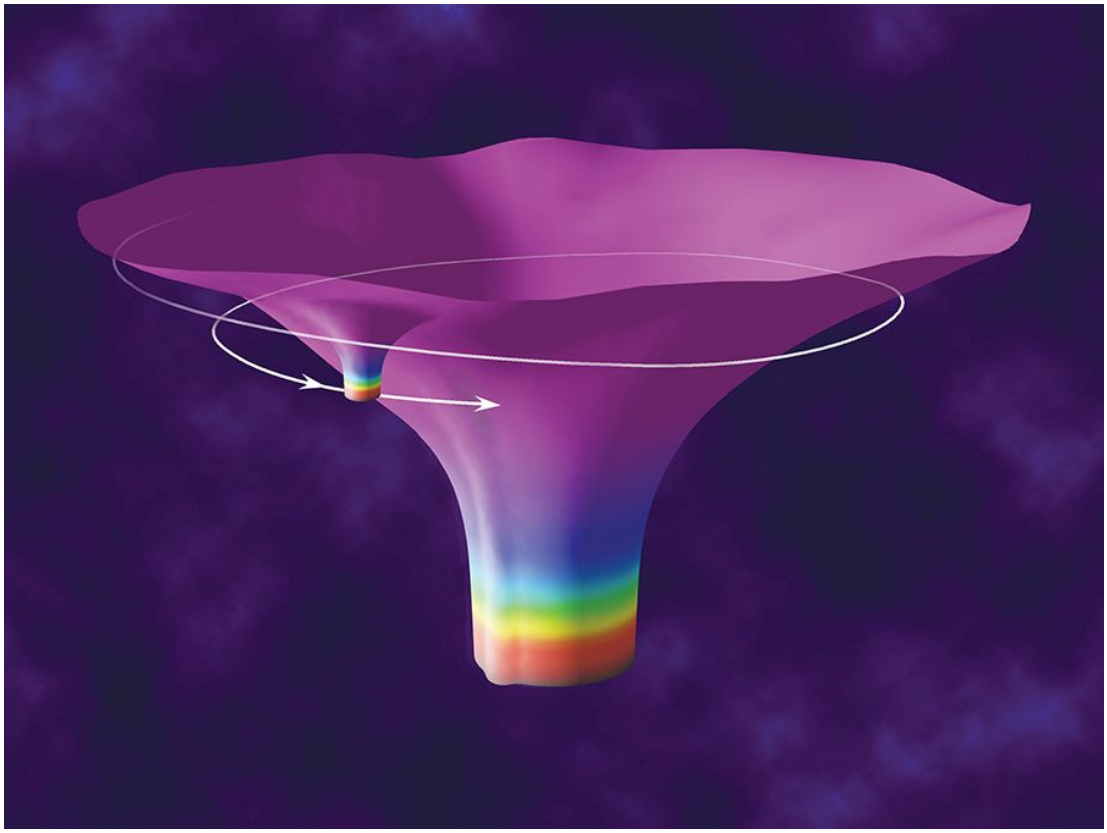


Fig. 21.1. A small black hole spiraling into a large black hole, as viewed from the bulk with one space dimension removed. *[Drawing by Don Davis based on a sketch by me.]*

Does the bulk really and truly exist, or is it just a figment of our imaginations? Until the 1980s, most physicists, including me, thought it a figment.

How could it be a figment? Don't we know for sure that our universe's space is warped? Don't the radio signals sent to the *Viking* spacecraft reveal its warpage to high precision (Chapter 4)? *Yes*. . . . And since our space is truly warped, doesn't it have to be warped inside some higher-dimensional space, inside some bulk?

No. It is perfectly possible for our universe to be warped without there really existing a higher-dimensional bulk. We physicists can describe our universe's warping, in mathematics, without the aid of a bulk. We can formulate Einstein's relativistic laws, which govern the warping, without the aid of a bulk. In fact, that's how we almost always